SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1703

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.* (BTL 5)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Design Task

Code of Conduct:

*I **Dhanajeyan J(192324211)**certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Design Task

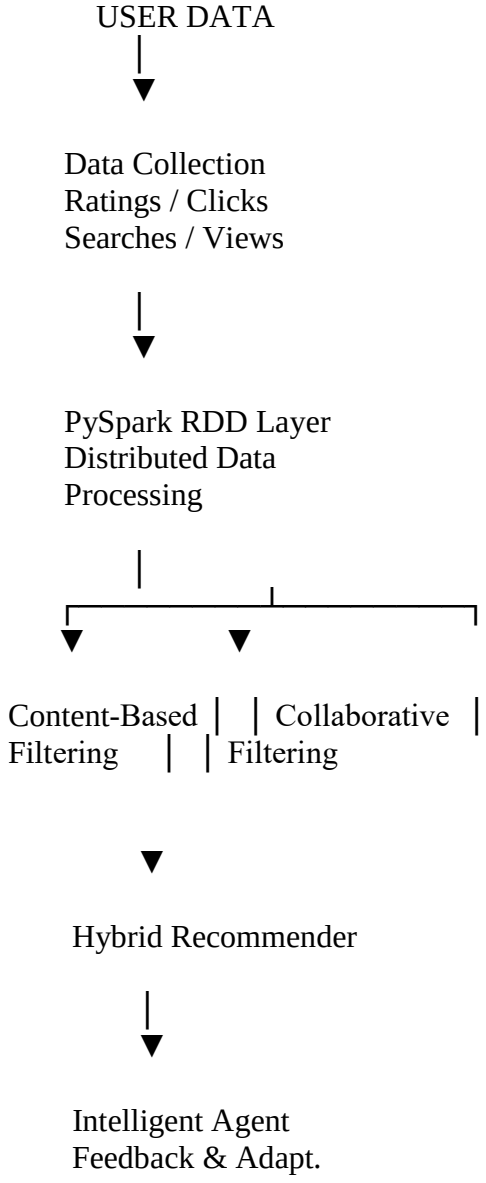
1. **Hybrid AI Recommendation Engine Design**

   Design a hybrid AI-based recommendation system that integrates content-based and collaborative filtering techniques using PySpark RDDs. Explain how distributed data processing can be used to handle large-scale user-item interactions, and describe the role of intelligent agents in adapting recommendations based on user behavior and feedback in real time.

Objective

Design a scalable recommendation system that combines **Content-Based Filtering** and **Collaborative Filtering** using **PySpark RDDs**. The system should analyze large-scale user-item interactions and continuously adapt recommendations according to user behavior and feedback.

System Architecture



Personalized Items

PySpark RDD Processing

RDDs can distribute user-item data across multiple worker nodes.

Example:

```
ratings = sc.textFile("ratings.csv")

data = ratings.map(lambda x: x.split(","))

user_items = data.map(
    lambda x: (x[0], (x[1], float(x[2])))
)

high_rated = user_items.filter(
    lambda x: x[1][1] >= 4
)
```

Common RDD operations include:

- map() – transform records
- filter() – remove unwanted records
- flatMap() – generate multiple records
- reduceByKey() – aggregate ratings
- groupByKey() – group interactions
- join() – combine user/item information
- sortBy() – rank recommendations

Recommendation Techniques

Content-Based Filtering

Recommends items similar to those previously liked by the user.

For example:

User liked:
→ Action Movies
→ Sci-Fi Movies

System recommends:
→ Similar Action/Sci-Fi movies

Collaborative Filtering

Uses the behavior of similar users.

User A → Movie 1, Movie 2

User B → Movie 1, Movie 2, Movie 3

Therefore:
Movie 3 → recommended to User A

Hybrid Approach

The final recommendation score can be calculated as:

Final Score =
α × Content Score +
(1-α) × Collaborative Score

where α controls the contribution of each technique.

Role of Intelligent Agents

An intelligent recommendation agent:

1. Monitors user behavior.
2. Collects clicks, ratings, searches and purchases.
3. Updates the user's preference profile.
4. Detects changes in interests.
5. Generates personalized recommendations.
6. Learns from positive/negative feedback.

For example:

User watches more AI videos
↓
Agent detects interest
↓
User profile updated
↓
AI-related content gets higher score
↓
New recommendations generated

Benefits

- Handles millions of interactions.
- Distributed processing reduces computation time.
- Combines two recommendation techniques.
- Provides personalized recommendations.
- Continuously learns from feedback.
- Scalable for large applications.

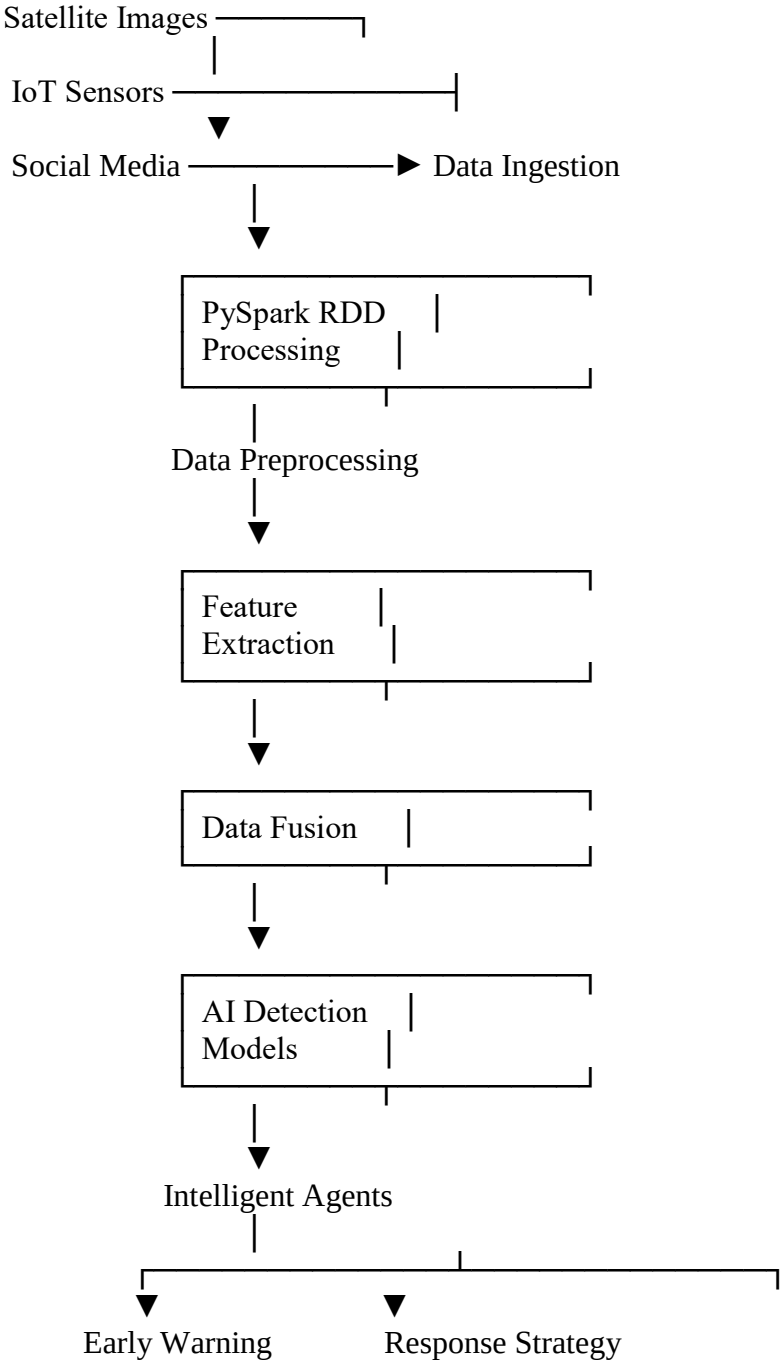
2. **Smart Disaster Detection System**

Develop a distributed AI system using PySpark and RDD operations to detect natural disasters (such as floods or earthquakes) from multi-source data streams (satellite images, sensors, and social media). Describe the system architecture, data fusion techniques, and how intelligent agents coordinate to provide early warnings and response strategies.

Objective

Develop a distributed AI system using **PySpark RDDs** to detect floods, earthquakes and other natural disasters from **satellite images, IoT sensors and social-media data**.

System Architecture



RDD-Based Processing

```
sensor_data = sc.textFile("sensor_data.csv")

records = sensor_data.map(
    lambda x: x.split(",")
)

valid_data = records.filter(
    lambda x: x[2] != "NULL"
)

water_levels = valid_data.map(
    lambda x: (x[0], float(x[2]))
)

average_levels = water_levels.reduceByKey(
    lambda a, b: (a + b) / 2
)
```

Data Fusion

Data fusion combines information from multiple sources.

Example:

```
Heavy Rainfall
      +
River Water Level Increasing
      +
Satellite Shows Water Expansion
      +
People Report Flooding
      ↓
High Flood Probability
```

Combining multiple sources improves reliability and reduces false alarms.

Intelligent Agents

Different agents can perform specialized tasks:

Sensor Agent

- Monitors sensor readings.
- Detects abnormal measurements.

Satellite Agent

- Analyzes satellite information.
- Identifies affected geographical regions.

Social Media Agent

- Extracts disaster-related posts.
- Identifies location and severity.

Warning Agent

- Calculates disaster risk.
- Sends alerts to authorities and citizens.

Response Agent

- Suggests evacuation routes.
- Identifies emergency resources.

Workflow

Data Collection
↓
RDD Distribution
↓
Data Cleaning
↓
Feature Extraction
↓
Multi-source Data Fusion
↓
AI Disaster Detection
↓
Risk Classification
↓
Agent Coordination
↓
Early Warning
↓
Emergency Response

Benefits

- Early disaster detection.
- Processes large-scale data.
- Combines multiple data sources.
- Reduces false alarms.
- Supports real-time decision making.
- Improves emergency response.

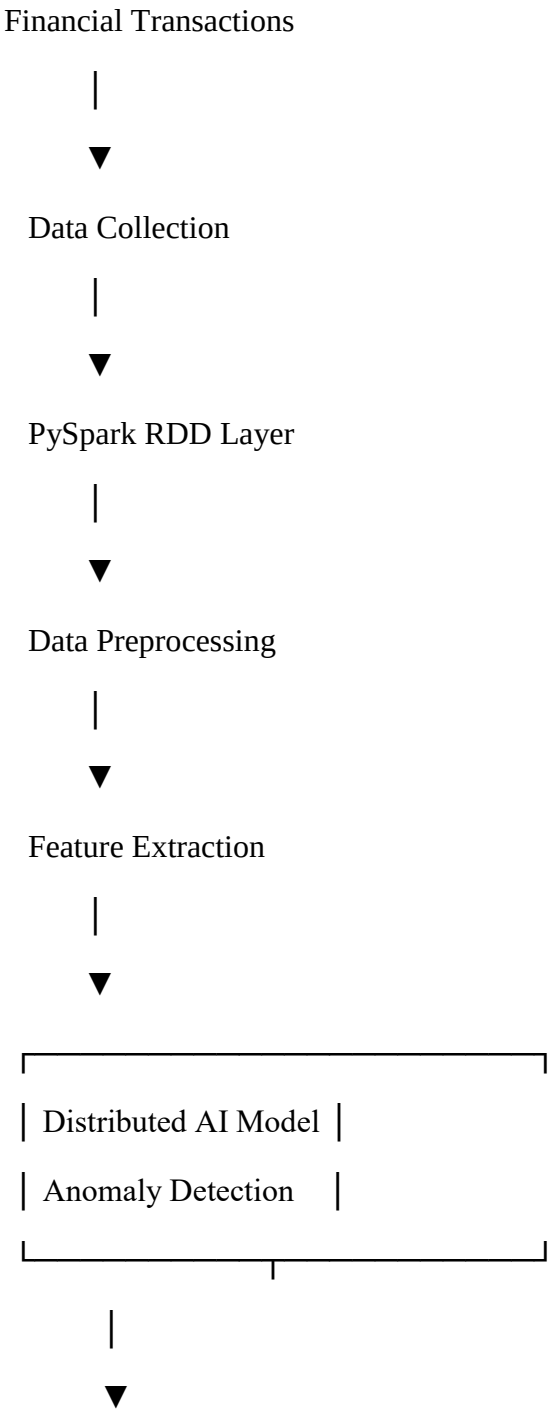
3. **AI-Based Fraud Detection Framework**

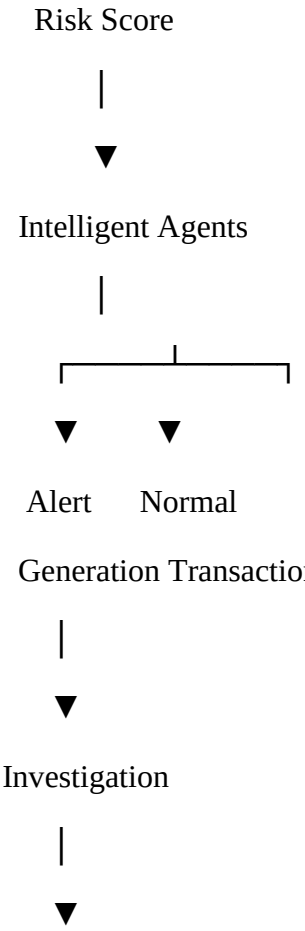
   Design a scalable fraud detection system using PySpark RDDs to process large volumes of financial transactions. Explain how anomaly detection algorithms can be implemented in a distributed manner and how intelligent agents collaborate to identify suspicious patterns, trigger alerts, and continuously improve detection accuracy.

Objective

Design a scalable fraud detection framework using **PySpark RDDs** to analyze millions of financial transactions and detect suspicious behavior.

System Architecture





RDD Processing

```
transactions = sc.textFile("transactions.csv")

data = transactions.map(
    lambda x: x.split(",")
)

large_transactions = data.filter(
    lambda x: float(x[2]) > 100000
)

user_transactions = large_transactions.map(
    lambda x: (x[1], float(x[2]))
)

total_amount = user_transactions.reduceByKey(
    lambda a, b: a + b
)
```

Anomaly Detection

The system can calculate features such as:

- Transaction amount.
- Transaction frequency.

- Location change.
- Unusual transaction time.
- Device changes.
- Multiple transactions in a short period.

Example:

Normal behavior:
User → ₹1,000–₹5,000 transactions

Suddenly:
User → ₹2,00,000 transaction
Location → Different country
Time → 3:00 AM

↓

High Anomaly Score
↓
Potential Fraud

Intelligent Agent Collaboration

Transaction Monitoring Agent

- Monitors incoming transactions.

Anomaly Detection Agent

- Calculates anomaly scores.

Risk Assessment Agent

- Assigns low, medium or high risk.

Alert Agent

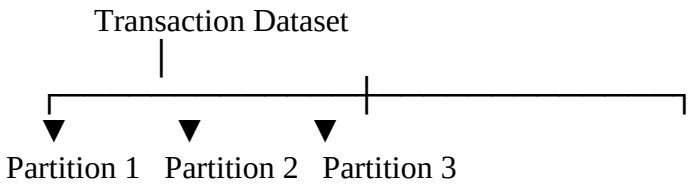
- Generates alerts for suspicious transactions.

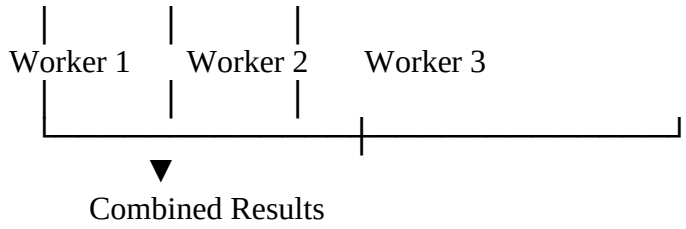
Learning Agent

- Uses confirmed fraud cases to improve future detection.

Distributed Processing

RDD partitions allow transactions to be processed simultaneously:





Benefits

- Handles massive transaction volumes.
- Fast distributed processing.
- Detects unusual patterns.
- Supports real-time alerts.
- Reduces financial losses.
- Continuously improves through feedback.

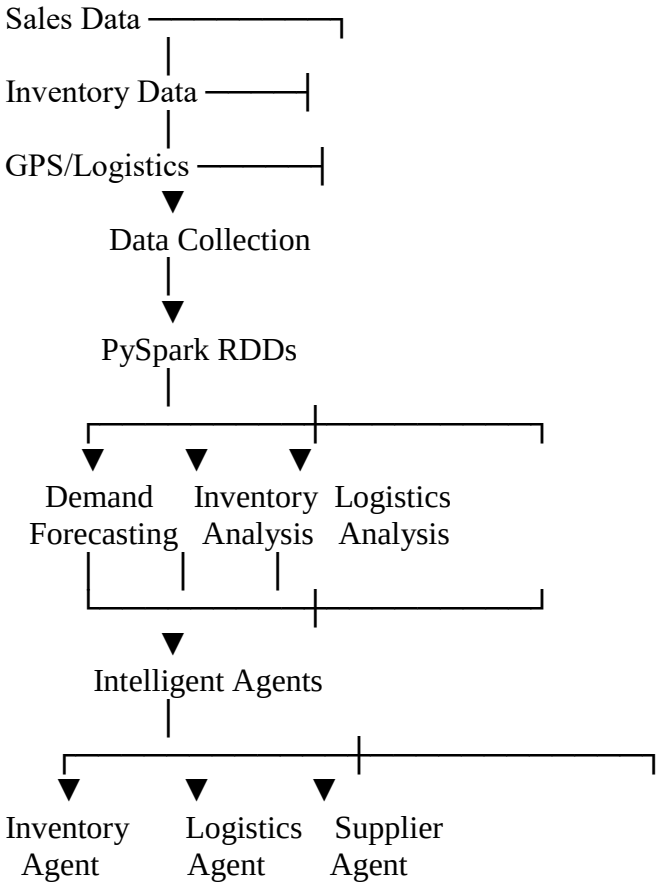
4. **Autonomous Supply Chain Optimization System**

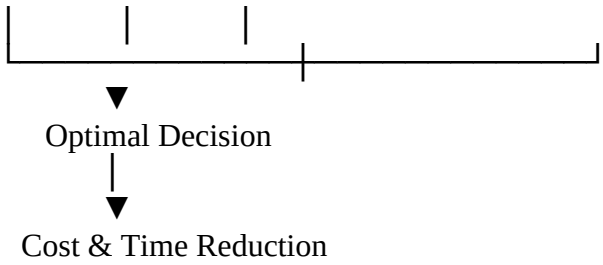
   Construct a distributed AI system using PySpark for optimizing supply chain operations (inventory, logistics, demand forecasting). Describe how RDD transformations support large-scale data processing and how multiple intelligent agents interact to make decentralized decisions for cost minimization and efficiency improvement.

Objective

Design a distributed AI system that optimizes **inventory, logistics and demand forecasting** using PySpark RDDs and multiple intelligent agents.

System Architecture





RDD Transformations

Suppose sales data contains:

Product ID, Quantity, Date, Location

RDD processing:

```
sales = sc.textFile("sales.csv")

records = sales.map(
   lambda x: x.split(",")
)

product_sales = records.map(
   lambda x: (x[0], int(x[1]))
)

total_sales = product_sales.reduceByKey(
   lambda a, b: a + b
)
```

Useful transformations include:

- map()
- filter()
- flatMap()
- reduceByKey()
- groupByKey()
- join()
- sortByKey()

Intelligent Agents

Inventory Agent

- Monitors stock.
- Predicts shortages.
- Recommends reorder quantities.

Demand Forecasting Agent

- Analyzes historical sales.
- Predicts future demand.

Logistics Agent

- Selects efficient transportation routes.
- Minimizes delivery time.

Supplier Agent

- Compares suppliers.
- Considers cost and delivery reliability.

Coordination Agent

- Combines decisions from different agents.

Decision Example

Demand Forecast
↓
Expected demand = 10,000 units
↓
Current inventory = 3,000 units
↓
Inventory Agent detects shortage
↓
Supplier Agent selects supplier
↓
Logistics Agent selects optimal route
↓
Order generated

Cost Optimization

The system attempts to minimize:

Total Cost =
Inventory Cost
+ Transportation Cost
+ Storage Cost
+ Ordering Cost
+ Delay Cost

Benefits

- Reduces inventory costs.
- Prevents stock-outs.
- Improves demand forecasting.
- Optimizes transportation.
- Supports decentralized decision-making.
- Handles large supply-chain datasets efficiently.

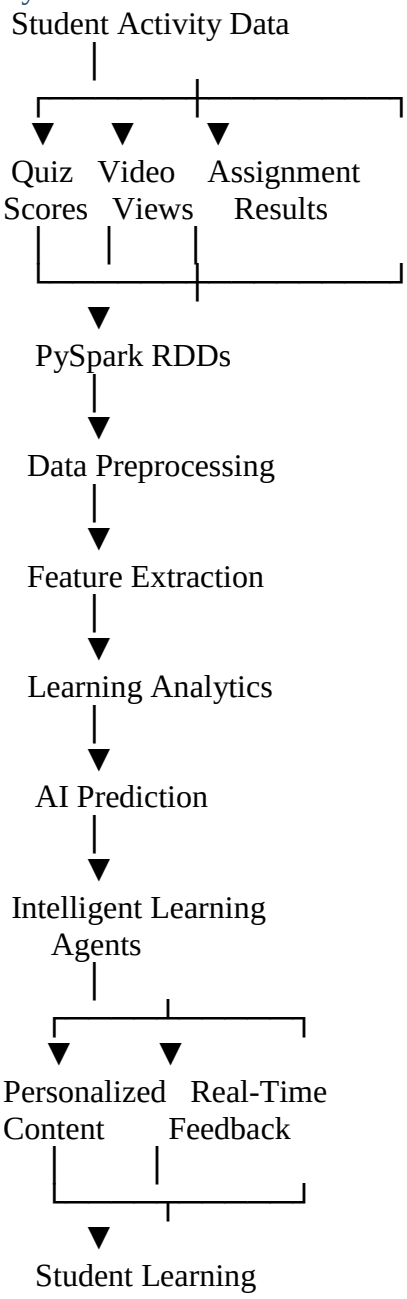
5. **Personalized Learning Analytics Platform**

   Design an AI-driven educational platform that uses PySpark and RDDs to analyze student learning behavior and performance data. Explain how the system supports adaptive learning through intelligent agents, real-time feedback, and scalable analytics to personalize content delivery for thousands of learners.

Objective

Develop an AI-based educational platform using **PySpark RDDs** to analyze student learning behavior and performance and provide personalized learning content.

System Architecture



Data Collected

The platform can analyze:

- Quiz scores.
- Assignment marks.
- Video watching time.
- Number of attempts.
- Login frequency.
- Topics completed.
- Questions answered incorrectly.
- Learning speed.

RDD Processing

```
students = sc.textFile("student_data.csv")

data = students.map(
   lambda x: x.split(",")
)

scores = data.map(
   lambda x: (x[0], float(x[2]))
)

average_scores = scores.combineByKey(
   lambda value: (value, 1),
   lambda acc, value: (acc[0] + value, acc[1] + 1),
   lambda acc1, acc2:
      (acc1[0] + acc2[0], acc1[1] + acc2[1])
)
```

RDDs distribute student records among different worker nodes, allowing thousands or millions of learning activities to be processed efficiently.

Intelligent Agents

Student Monitoring Agent

- Tracks student activity.

Performance Agent

- Analyzes marks and learning progress.

Content Recommendation Agent

- Selects appropriate learning materials.

Feedback Agent

- Provides immediate feedback.

Difficulty Adjustment Agent

- Increases or decreases question difficulty based on performance.

Adaptive Learning Example

Student performs poorly in DBMS normalization
↓
Performance Agent detects weakness
↓
Recommendation Agent selects
basic normalization material
↓
Student completes practice questions
↓
Performance improves
↓
Agent increases difficulty

Personalization

Students can be classified according to their learning behavior:

Student Type	Recommended Action
High Performer	Advanced problems
Average Performer	Regular practice
Weak Performer	Basic concepts + tutorials
Inactive Student	Reminders + engagement content
Fast Learner	Higher difficulty

Real-Time Feedback

The platform continuously monitors student activity:

Student Activity
↓
RDD Processing
↓
Performance Analysis
↓
Agent Decision
↓
Personalized Feedback
↓
Updated Student Profile

Benefits

- Supports thousands of students.
- Provides personalized learning.
- Detects weak topics.
- Gives immediate feedback.
- Dynamically adjusts difficulty.
- Improves student engagement.
- Enables data-driven educational decisions.

RUBRICS FOR CALCULATION

Evaluation Criteria	Problem Understanding & Requirement Analysis	System Architecture Design	Use of PySpark & RDD Operations	Intelligent Agent Integration	Scalability & Distributed Computing Design	Data Flow & Processing Pipeline	Innovation & Creativity	Clarity, Presentation & Documentation
Weightage	10%	20%	15%	15%	10%	10%	10%	10%

Criteria	Excellent (5)	Good (4)	Average (3)	Poor (2)
Problem Understanding & Requirement Analysis	Clearly defines problem, objectives, and constraints	Mostly clear with minor gaps	Basic understanding	Unclear or incorrect
System Architecture Design	Complete, scalable, well-structured architecture with diagrams	Good design with minor issues	Basic design, lacks detail	Poor/incomplete
Use of PySpark & RDD Operations	Efficient and correct use of RDD transformations/actions	Mostly correct usage	Limited or partial use	Incorrect/missing
Intelligent Agent Integration	Strong integration with clear agent roles and logic	Good integration	Basic integration	Poor/no integration
Scalability & Distributed Computing Design	Clearly addresses scalability, fault tolerance, parallelism	Covers most aspects	Limited consideration	Not addressed
Data Flow & Processing Pipeline	Clear, logical, and well-defined data flow	Mostly clear	Some gaps	Poorly defined

Innovation & Creativity	Highly innovative, realistic, and impactful solution	Some innovation	Basic idea	No originality
Clarity, Presentation & Documentation	Very clear, neat diagrams, well-organized report	Mostly clear	Somewhat organized	Poor presentation